

DANIELLA BAR-LEV, PH.D.

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PROFESSIONAL SUMMARY

Algorithm researcher and computer scientist. Ph.D. in Computer Science, two separate B.Sc. degrees in Computer Science and Mathematics (Magna cum laude). Proven expertise in coding theory, algorithms, and combinatorial structures. Experienced in designing coding schemes and algorithms for storage, with applications in reliable, high-density DNA-based data storage and cryptography systems. Demonstrated ability to translate theoretical research into practical solutions for data-intensive computational challenges, including float data compression and algorithmic modifications to graph indices. Mentored dozens of undergraduate and graduate projects, several resulting in publications, one of which won an ISITA Best Paper Award. Guest editor for an IEEE special issue and reviewer for top journals, including IEEE Trans. Information Theory (IEEE TIT), Nature Communications, and Science Advances.

SELECTED ACHIEVEMENTS

2026	Feder Family Prize (1st Place) for Outstanding Research in the Field of Communication Technologies.
2024	The Eric and Wendy Schmidt Postdoctoral Award for Women in Mathematical and Computing Sciences.
2024	VATAT Scholarship for Outstanding Postdoctoral Women Students.
2023	Blavatnik Prize for Outstanding Israeli Ph.D. Students in Computer Science.
2015 – 2019	Lapidim CS Excellence Program Scholar , the Technion's highly competitive undergraduate excellence track.

TECHNICAL SKILLS

Expertise:	Algorithms, Combinatorial Structures, Coding Theory, Discrete Probability
Domains:	Channel Coding, DNA-Based Data Storage, DNA-Based Cryptography (e.g., PIR, Secret Sharing), Bioinformatics, Invertible Bloom Lookup Tables (IBLT), Data Compression, Graph Indices
Programming:	Extensive experience in C and Python, working knowledge of other languages

EXPERIENCE

Senior Project Scientist <i>Technion – Israel Institute of Technology</i>	<i>Jun 2026 – Present</i>
Leading the development of novel algorithmic methodologies for DNA-based cryptography within the DiDAX project. Focusing on high-density encoding, secret sharing and secure information retrieval.	
Postdoctoral Research Scholar & Lecturer <i>Universität Zürich (UZH)</i>	<i>Jan 2026 – Present</i>
- Investigating Coded Information Retrieval and the fundamental limits of block-structured retrieval in DNA-based data storage. - Designing curriculum and lecturing class & seminar on coding for DNA-based information systems; TA in <i>Finite Fields</i>. - Advising M.Sc. thesis on reconstruction for synchronization channels.	
Postdoctoral Research Scholar <i>University of California, San Diego (UCSD)</i>	<i>Jan 2025 – Dec 2025</i>
- Developed an additive PTAS for dyadic codings of LLM outputs, providing a framework for LLM-based steganography. - Investigated the Labeled Coupon Collector Problem and generalized separating systems; mentored students on this topic.	
Lecturer <i>Reichman University, IDC Herzliya</i>	<i>2023 – 2024</i>
Lecturer for <i>Coding Theory</i> and <i>Logic and Set Theory</i> (150+ students); TA for <i>Computational Models</i>.	
Algorithmic Research Intern <i>Pinecone</i>	<i>Summer 2022</i>
Designed and developed dynamic deletion methods for graph indices such as HNSW, ensuring high information retrieval quality during real-time updates.	
Research Intern <i>VAST Data</i>	<i>Summer 2020</i>
Researched float data compression for spatio-temporal measurements; designed a compression tool utilizing vector operations for high-speed CPU performance.	
Intern Software Engineer <i>ScaleIO EMC (Dell)</i>	<i>2016 – 2018</i>
Designed, optimized, and implemented core features within the primary enterprise storage architecture.	

EDUCATION

Ph.D. in Computer Science <i>Technion – Israel Institute of Technology</i>	<i>2020 – 2024</i>
<i>Thesis:</i> "Theory and Practice of DNA Storage" <i>Advisors:</i> Prof. Tuvi Etzion & Prof. Eitan Yaakobi	
M.Sc. in Computer Science <i>Technion – Israel Institute of Technology</i>	<i>2019 – 2020</i>
B.Sc. in Computer Science & B.Sc. in Mathematics (Magna cum laude) <i>Technion – Israel Institute of Technology</i>	<i>2014 – 2019</i>

PATENTS & SELECTED PUBLICATIONS

- Patent Application:** Reconstruction of Information Stored in a DNA Storage System, US patent application No. 18/233,855 (2024).
- Publication:** "Scalable and robust DNA-based storage via coding theory and deep learning," *Nature Machine Intelligence*, Feb 2025.
- Publication:** "Cover your bases: how to minimize the sequencing coverage in DNA storage systems," *IEEE TIT*, Jan. 2025.
- Publication:** "The zettabyte era is in our DNA," *Nature Computational Science*, Nov 2024.